

# Ensemble-Based Identity Document Fraud Detection for the FREUID Challenge 2026

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**Weights:** <https://www.kaggle.com/datasets/ypsathore/freuid-2026-pratap424-weights>

## 1. Approach overview

We treat fraud detection as binary image classification with an ensemble designed to maximize recall of fraudulent documents at the strict APCER@1%BPCER operating point of the FREUID metric.

**Models (all trained June 30 – July 2, 2026, before the code freeze):**

| Backbone                                | Input | Folds      | Pretraining     |
|-----------------------------------------|-------|------------|-----------------|
| ConvNeXt-Small ( fb_in22k_ft_in1k_384 ) | 384px | 5          | ImageNet-22k→1k |
| EVA02-Large ( mim_m38m_ft_in22k_in1k )  | 448px | 1 (fold o) | MIM M38M        |

An EfficientNet-B3 (5 folds) was also trained but **excluded** after out-of-fold analysis showed it degraded the ensemble (OOF FREUID 0.107 vs 0.00008 for ConvNeXt; public LB confirmed: 0.288 alone).

## 2. Training

Stratified 5-fold CV by label × document type (10 strata)

Loss: 0.5·BCE + 0.5·Focal ( $\gamma=2$ ,  $\alpha=0.75$ ), label smoothing 0.1

AdamW, cosine schedule with 2-epoch warmup, AMP, EMA (decay 0.9998)

Mixup ( $\alpha=0.4$ ) / CutMix ( $\alpha=1.0$ ) at  $p=0.5$

Heavy augmentation targeting capture-condition shift: geometric (RRC, flip, shift-scale-rotate, perspective), photometric (color jitter, brightness/contrast), degradation (Gaussian noise, JPEG compression 30–95, downscale 0.4–0.9, blur), occlusion (coarse dropout). A silent albumentations-v2 API incompatibility initially disabled 4 of these; fixing it improved public LB from 0.144 to 0.133 (single fold).

## 3. Inference and ensembling

8-view TTA on public test; staged TTA on private test (identity + horizontal flip)

All model×fold×TTA predictions aggregated as:  $\alpha \cdot \text{mean} + (1-\alpha) \cdot \text{max}$  with  $\alpha=0.5$ .

The max term preserves fraud evidence found by any single ensemble member — at the 1% BPCER operating point, a missed fraud is far costlier than a slightly elevated bona-fide score. Public LB validated this:  $\alpha=0.5$  (0.07894) >  $\alpha=0.7$  (0.07955) > plain mean (0.08022) > power-mean  $p=3$  (0.07974) >> rank averaging (0.12695).

Score calibration (isotonic/Platt) was evaluated and rejected: both FREUID components are ranking-based, so monotonic recalibration cannot change the score.

## 4. Results

| Submission                                                      | Public LB                      |
|-----------------------------------------------------------------|--------------------------------|
| EfficientNet-B3 fo (excluded model)                             | 0.28818                        |
| ConvNeXt-Small fo                                               | 0.13277                        |
| ConvNeXt-Small 5-fold                                           | 0.10950                        |
| + TTA                                                           | 0.09786                        |
| + EVA02-Large fo                                                | 0.08237                        |
| + 8-view TTA everywhere, maxblend $\alpha=0.7$                  | 0.07955                        |
| + <b>maxblend <math>\alpha=0.5</math> (final public config)</b> | <b>0.07894</b>                 |
| Full submission incl. real private predictions (v1, v2)         | 0.07894 public; private hidden |

Final submissions predict all 142,818 test rows with model outputs (no placeholder values). Private-test inference used the identical frozen weights with staged TTA (1, 2, then 8 views).

## 5. Code-freeze compliance

All weights predate the July 13 private-test release (training logs and file timestamps available for audit). Post-freeze work consisted exclusively of: inference on private images with unchanged weights, ensemble aggregation, documentation, and packaging. A fold-1 EVA02 run accidentally launched on July 13 was terminated mid-run and its checkpoint deleted; it contributes to nothing.

## 6. Reproducibility

`README.md` — full pipeline instructions

`Dockerfile` — pinned environment

Hardware: single NVIDIA RTX A5000 (24GB); total training  $\approx$  60 GPU-hours